

```
frequency_penalty=0.5,
presence_penalty=0.5
).choices[0].text
# Interface with Google search API to retrieve relevant articles
url = "https://www.googleapis.com/customsearch/v1"
params = {
    "q": keywords,
    "num": 5,
    "cx": "your-search-engine-id",
    "key": "your-api-key"
}
response = requests.get(url, params=params)
# Use PageRank algorithm to rank the articles based
on relevance
articles = response.json()["items"]
for article in articles:
    article["rank"] = pagerank(article["link"])
# Select the top 5 articles and format the response
top_articles = sorted(articles, key=lambda x: x["rank"],
reverse=True)[:5]
response = f"The top articles on machine learning are:\n"
for i, article in enumerate(top_articles):
    response += f"{i+1}. {article['title']}\n{article['link']}\n\n"
print(response)
```

7. Automatic Prompt Engineer (APE)

Automatic prompt engineer (APE) is a framework for automatic instruction creation and selection proposed by Zhou et al. in 2022. The issue of instruction creation is presented as a

black-box optimisation problem involving natural language synthesis, and LLMs are used to create and search through potential solutions.

In the first stage, output demonstrations are submitted to a big language model (as an inference model) to create instruction candidates for a job. The search process will be guided by these potential solutions. Using a target model to execute the instructions, the best instruction is then chosen based on computed evaluation scores.

The real potential of language models may be unlocked, and their reasoning abilities can be improved, through the use of sophisticated prompting mechanisms. These approaches enable users to develop more precise, powerful, and sophisticated prompts that tackle challenging tasks and difficulties by utilising techniques like few-shot prompting, chain of thought prompting, self-consistency, knowledge creation, and autonomous prompt engineering. These cutting-edge methods enhance the language model's capacity to provide contextually appropriate replies while also allowing it to demonstrate more subtle and cogent thinking. As a result, real-world issues are tackled with more flexibility and accuracy, bridging the gap between artificial intelligence and human-like cognition.

The future of AI-driven problem-solving will be shaped by improved prompting approaches as we continue to push the limits of language models and artificial intelligence. These methods, which emphasise the value of user-centric design and real-time adaptability, have the potential to change how we engage with technology and reshape our perceptions of what AI is capable of. 